

Project Initialization and Planning Phase

Date	28 JUNE 2026
Team ID	
Project Title	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve agricultural productivity using Machine Learning by recommending the most suitable crop based on soil nutrients and environmental conditions. The system analyzes parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to generate accurate crop recommendations. This helps farmers make informed decisions, optimize resource utilization, and promote sustainable farming practices.

Project Overview	
Objective	The primary objective is to develop an intelligent crop recommendation system that uses Machine Learning to analyze soil and environmental conditions, helping farmers select the most suitable crop for higher productivity and sustainable agriculture.
Scope	The project focuses on collecting agricultural data, preprocessing it, training a Machine Learning model, and deploying a Flask-based web application that provides real-time crop recommendations for farmers and agricultural stakeholders.
Problem Statement	
Description	Farmers often face challenges in selecting suitable crops due to changing climate conditions, lack of soil analysis, and limited access to data-driven agricultural guidance. This results in poor crop yield and inefficient use of resources.
Impact	Solving this problem improves crop productivity, reduces farming risks, optimizes fertilizer and water usage, supports sustainable agriculture, and helps farmers make informed decisions.
Proposed Solution	
Approach	Develop a Machine Learning-based crop recommendation system that processes soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall. The trained model predicts the most suitable crop and displays the recommendation through a Flask web application.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3 or above
Memory	RAM specifications	Minimum 4 GB RAM
Storage	Disk space for data, models, and logs	Minimum 10 GB free disk space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, Anaconda Navigator, Visual Studio Code / PyCharm
Data		
Data	Source, size, format	Crop Recommendation Dataset (CSV containing N, P, K, temperature, humidity, pH, rainfall, and crop labels)